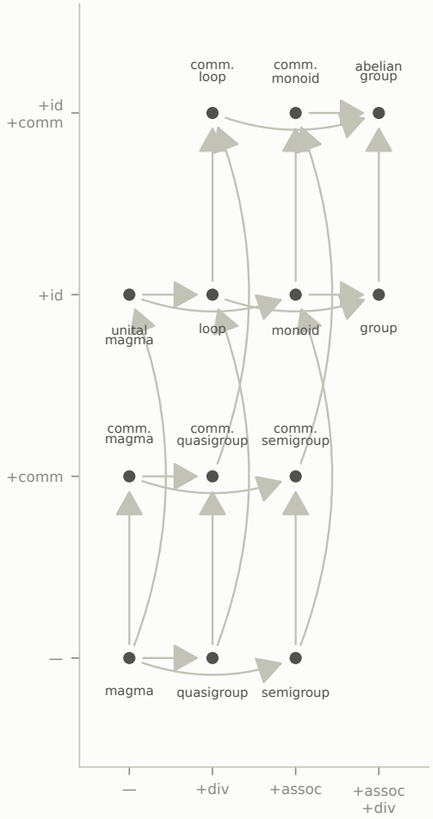
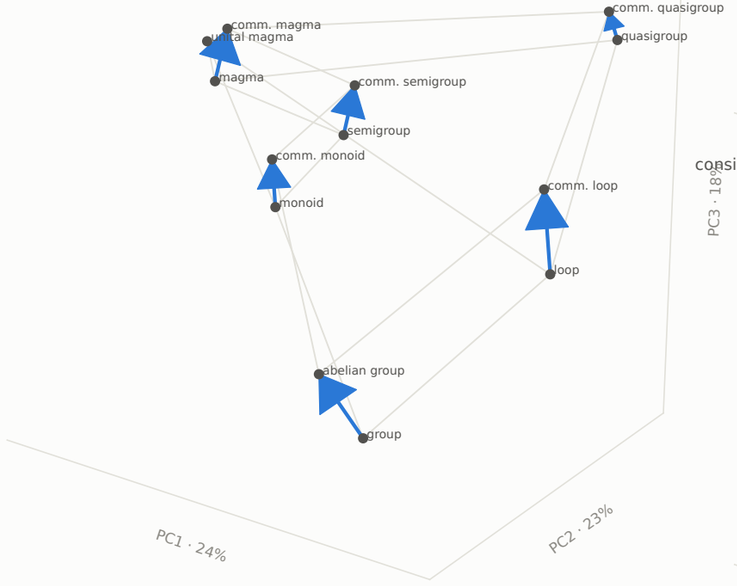
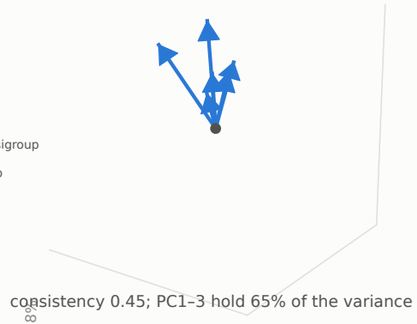
1 · Structures are sets of axioms



2 · Adding an axiom is an arrow + commutativity in Qwen2.5-Math-1.5B (top-3 PCs)

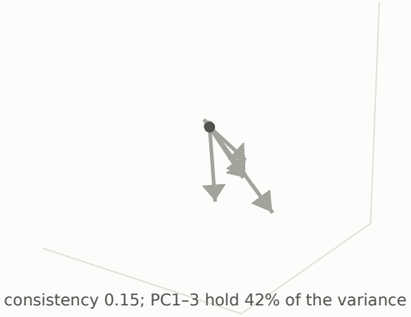


3 · Line the arrows up



consistency 0.45; PC1–3 hold 65% of the variance

random-init twin

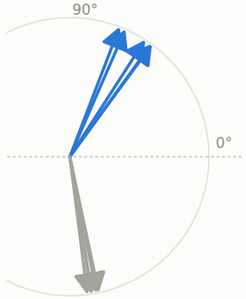


consistency 0.15; PC1–3 hold 42% of the variance

4 · Measure in the full space: angle of each arrow to the others' mean

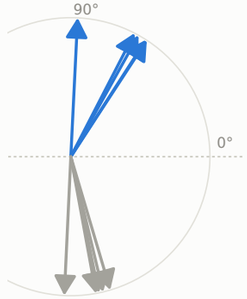
model above the dashed line (blue), twin below (grey); 0° = same direction, 90° = unrelated

+ associativity



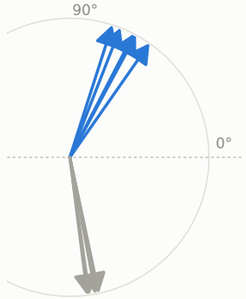
consistency 0.35 (p = 0.009)
twin 0.09

+ identity



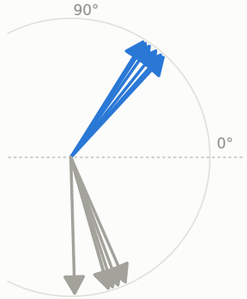
consistency 0.27 (p = 0.026)
twin 0.09

+ divisibility



consistency 0.30 (p = 0.019)
twin 0.10

+ commutativity



consistency 0.45 (p < 0.001)
twin 0.15